

§11.1 - Sequences

Definition #1 (Sequence)

A **sequence** (a_n) is an ordered list of numbers of the form

$$(a_1, a_2, a_3, \dots, a_n, a_{n+1}, \dots).$$

The book uses $\{a_n\}$ to denote a sequence.

We have

$$a_n = n$$

generating the natural numbers,

$$(b_n) = (0, 1, -1, 2, -2, \dots)$$

generating the integers.

While

$$(c_n) = (2, 1, 3, 4, \dots)$$

is not equal to (a_n) since

$$1 = a_1 \neq c_1 = 2.$$

Two sequences are equal if all of their terms are the same,

$$a_n = b_n$$

for all n .

Example #1 (Explicit Sequence)

List out the first three terms of

$$a_n = \frac{1}{3^n} \quad \text{and} \quad b_n = \frac{(-1)^n n}{n^2 + 1}.$$

Here

$$a_1 = \frac{1}{3}$$

$$a_2 = \frac{1}{9}$$

$$a_3 = \frac{1}{27}$$

$$b_1 = \frac{-1}{1+1} = -\frac{1}{2}$$

$$b_2 = \frac{1 \cdot 2}{4+1} = \frac{2}{5}$$

$$b_3 = \frac{-1 \cdot 3}{9+1} = -\frac{3}{10}$$

Example #2 (Recursive / Implicit Sequence)

If

$$c_{n+1} = 2c_n + 1 \quad \text{and} \quad c_1 = 1$$

find c_3 .

$$c_1 = 1$$

$$c_2 = 2c_1 + 1 = 2 + 1 = 3$$

$$c_3 = 2c_2 + 1 = 2 \cdot 3 + 1 = 7$$

In general,

$$c_n = 2^n - 1,$$

and you can see the proof in Math 55.

Informal Definition #1 (Convergence)

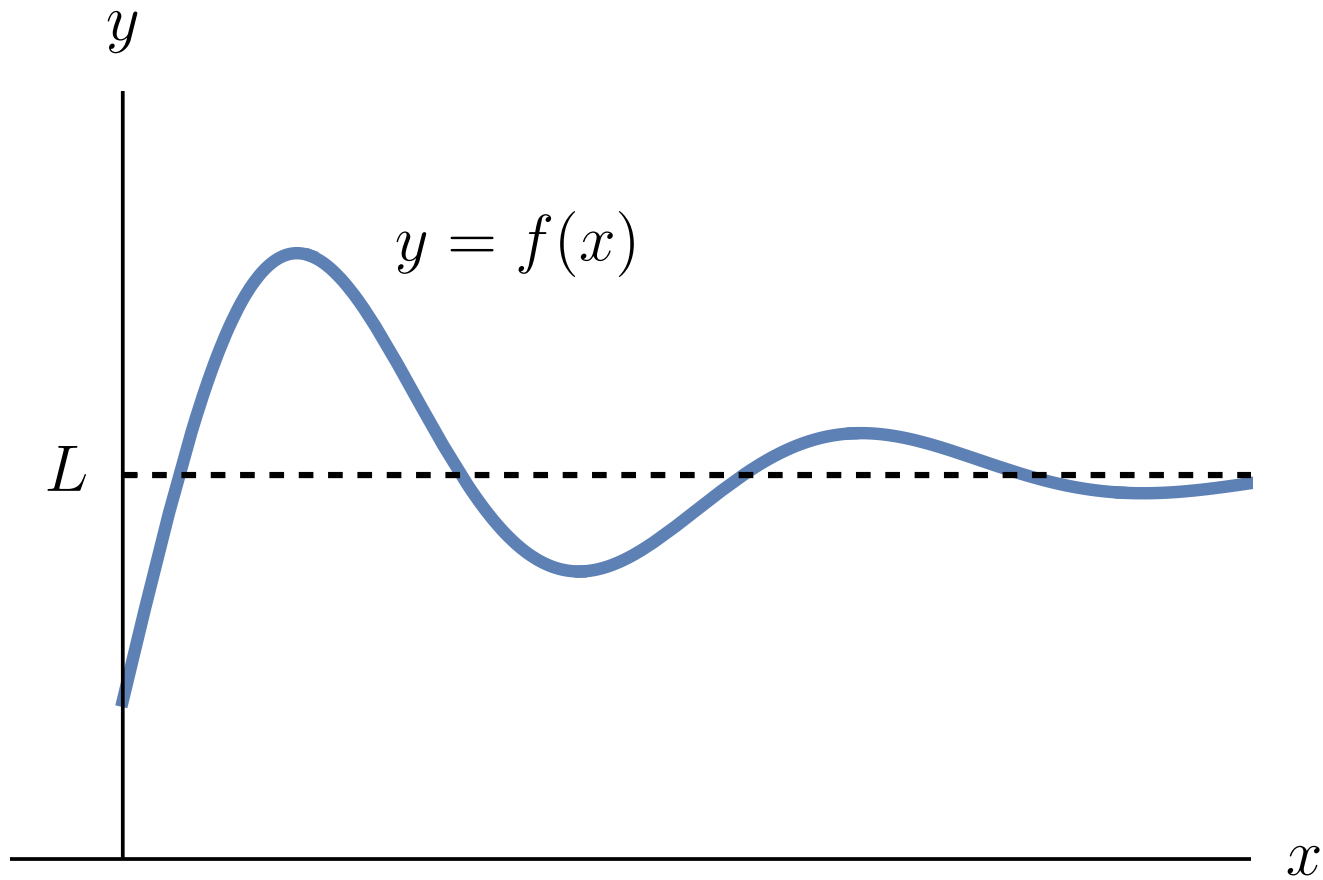
Let (a_n) be a sequence and $L \in \mathbf{R}$.

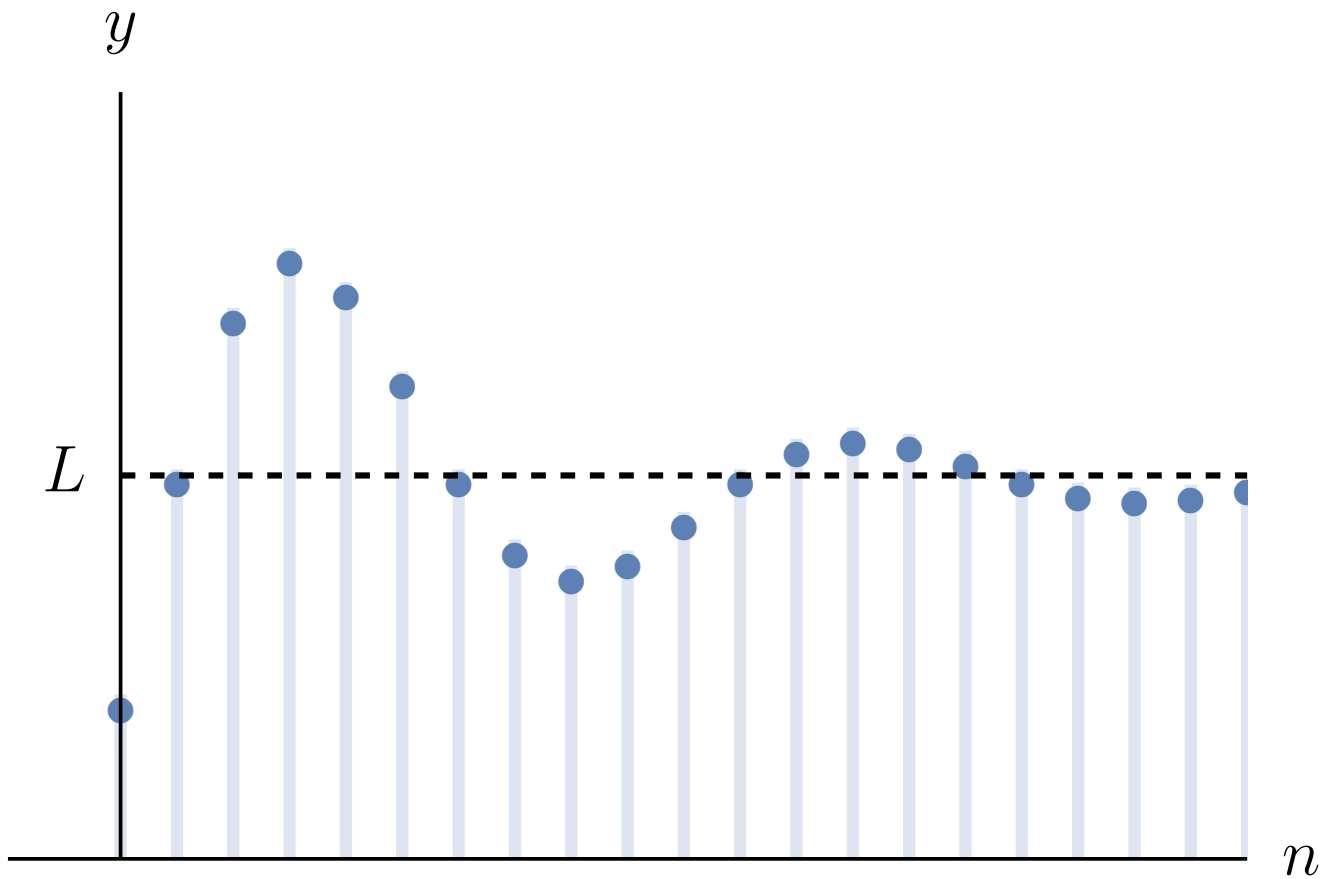
If we can make the terms a_n as close to L as we like by taking n sufficiently large then L is the **limit** of (a_n) .

We denote this by

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L$$

and this can also be expressed as (a_n) **converges** to L .





We borrow the same ideas behind limits of functions:

$$\lim_{x \rightarrow \infty} f(x) = L$$

means

$$f(x) \approx L$$

for all x "large". Therefore

$$a_n \rightarrow L$$

means

$$a_n \approx L$$

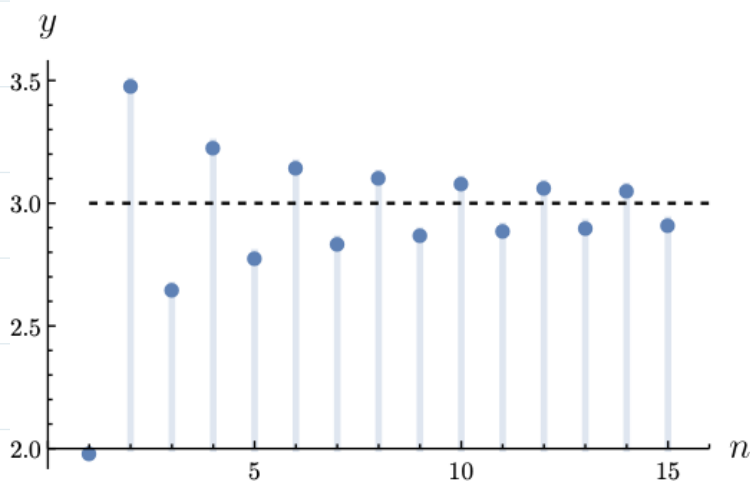
for all n big. In broad strokes, the terms a_n are clustering near L .

If

$$a_n = 3 + \frac{(-1)^n}{n}$$

then

$$a_n \rightarrow 3.$$



Definition #2 (Divergence)

Let (a_n) be a sequence.

If

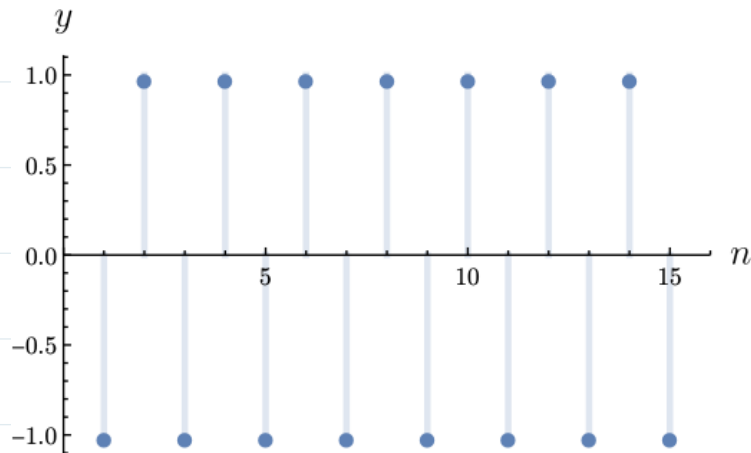
$$\lim_{n \rightarrow \infty} a_n$$

does not exist then (a_n) **diverges**.

What causes divergence:

1. ∞ -many terms in (a_n) are clustered near two (or more) distinct points in $\mathbb{R}$:

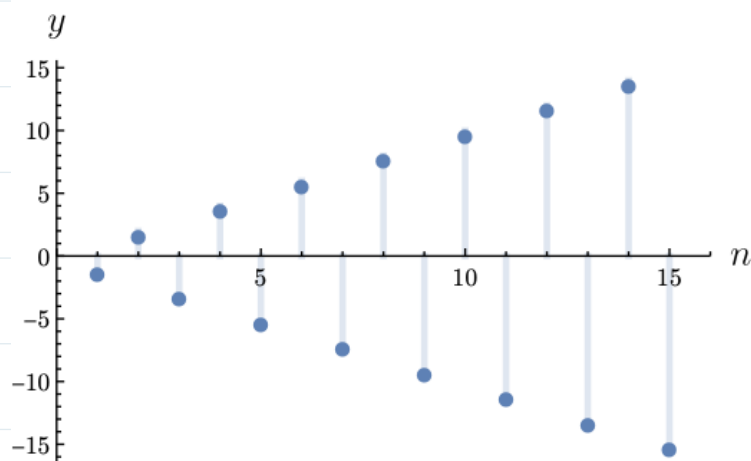
$$(a_n) = (1, -1, 1, -1, \dots).$$



2. ∞ -many terms in (a_n) grow without bound:

$$a_n = n \quad \text{or} \quad a_n = (-1)^n n$$

So ∞ -many terms of $(|a_n|)$ grow large.



Definition #3 (Equality)

Let $x, y \in \mathbf{R}$.

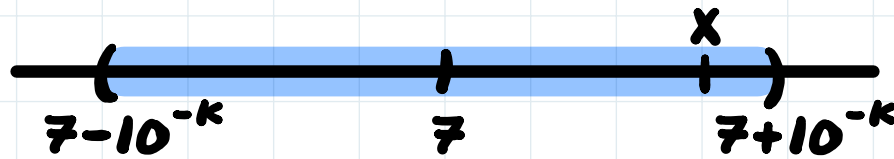
The following are equivalent

1. $x = y$

2. for all $\epsilon > 0$ we have

$$|x - y| < \epsilon.$$

We want to think of ε as being very small.



Assume

$$|x - 7| < 10^{-k} \quad *$$

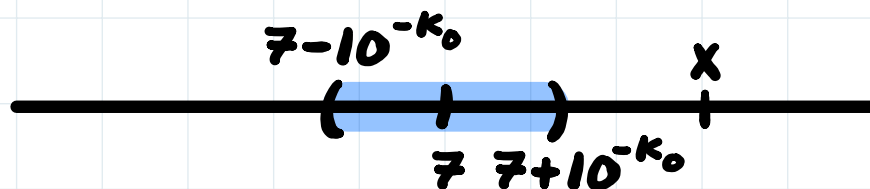
for all $k > 0$. This means that x and 7 are infinitely close to each other.

As $k \rightarrow \infty$,

$$|x - 7| \rightarrow 0.$$

If $x \neq 7$, then there is distance between those two numbers,

$$|x - 7| > 0.$$



We can then choose k_0 large such that

$$|x - 7| \geq 10^{-k_0}$$

but this violates $*$! Thus $x = 7$.

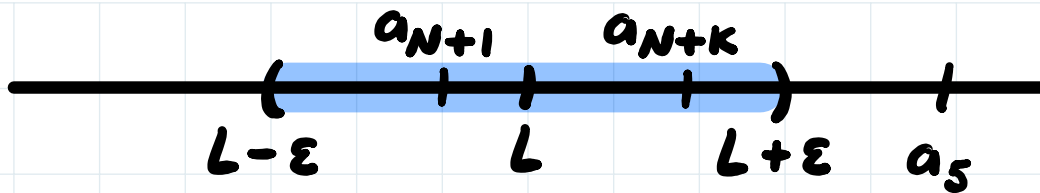
Definition #4 (Limit)

Let (a_n) be a sequence and $L \in \mathbf{R}$.

If for every $\epsilon > 0$ we have an index $N \in \mathbf{N}$ such that

$$n > N \quad \text{implies} \quad |a_n - L| < \epsilon$$

then (a_n) converges to L .



For any $\epsilon > 0$, all but finitely many a_n 's are in the interval

$$(L - \epsilon, L + \epsilon)$$

and it does not matter how small the interval is!

Theorem #1 (Basic Limit Laws)

Assume

$$a_n \rightarrow a \quad \text{and} \quad b_n \rightarrow b.$$

Then

1. $1/n \rightarrow 0$
2. $a_n + b_n \rightarrow a + b$
3. $ka_n \rightarrow ka$, for $k \in \mathbf{R}$
4. $a_nb_n \rightarrow ab$
5. $\frac{a_n}{b_n} \rightarrow \frac{a}{b}$ provided $b \neq 0$
6. $a_n \rightarrow 0$ if and only if $|a_n| \rightarrow 0$.

Example #3

Find

$$\lim_{n \rightarrow \infty} \sqrt{\frac{1 + 4n^2}{1 + n^2}}.$$

We can reuse ideas from Math 51, and multiply all terms by n^{-p} where p is the highest power of n in

$$\left(\frac{1+4n^2}{1+n^2} \right)^{1/2} \Rightarrow p = 2.$$

Then

$$\left(\frac{1+4n^2}{1+n^2} \right)^{1/2} = \left(\frac{1/n^2 + 4}{1/n^2 + 1} \right)^{1/2} \rightarrow \frac{4^{1/2}}{1} = 2.$$

Our Limit Laws imply

$$\lim_{n \rightarrow \infty} \left(\frac{1+4n^2}{1+n^2} \right)^{1/2} = 2.$$

Theorem #2 (Squeeze Theorem)

Suppose

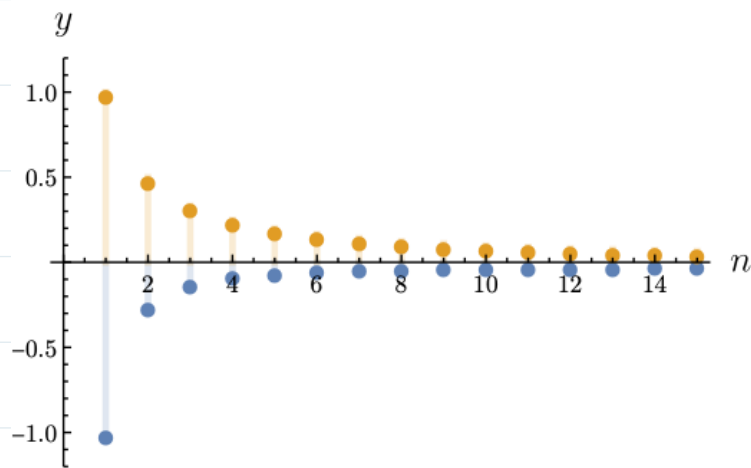
$$a_n \rightarrow L \quad \text{and} \quad c_n \rightarrow L.$$

If (b_n) satisfies

$$a_n \leq b_n \leq c_n$$

for all $n \geq n_0$ then

$$b_n \rightarrow L.$$



If

$$-\frac{1}{n^2} < b_n \leq \frac{1}{n}$$

then $b_n \rightarrow 0$ by the Squeeze Theorem.

The Theorem works in a similar manner as the Comparison Test for improper integrals.

Theorem #3

If $a_n \rightarrow L$ and f is continuous at L then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L).$$

If $a_n \rightarrow 3$ and $f(x) = x^2 + e^x$ then

$$\mathbf{f(a_n) = a_n^2 + e^{a_n} \rightarrow 3^2 + e^3}$$

as $n \rightarrow \infty$.

Definition #5 (Increasing/Decreasing)

If

$$a_n < a_{n+1}$$

for all n then (a_n) is an **increasing** sequence.

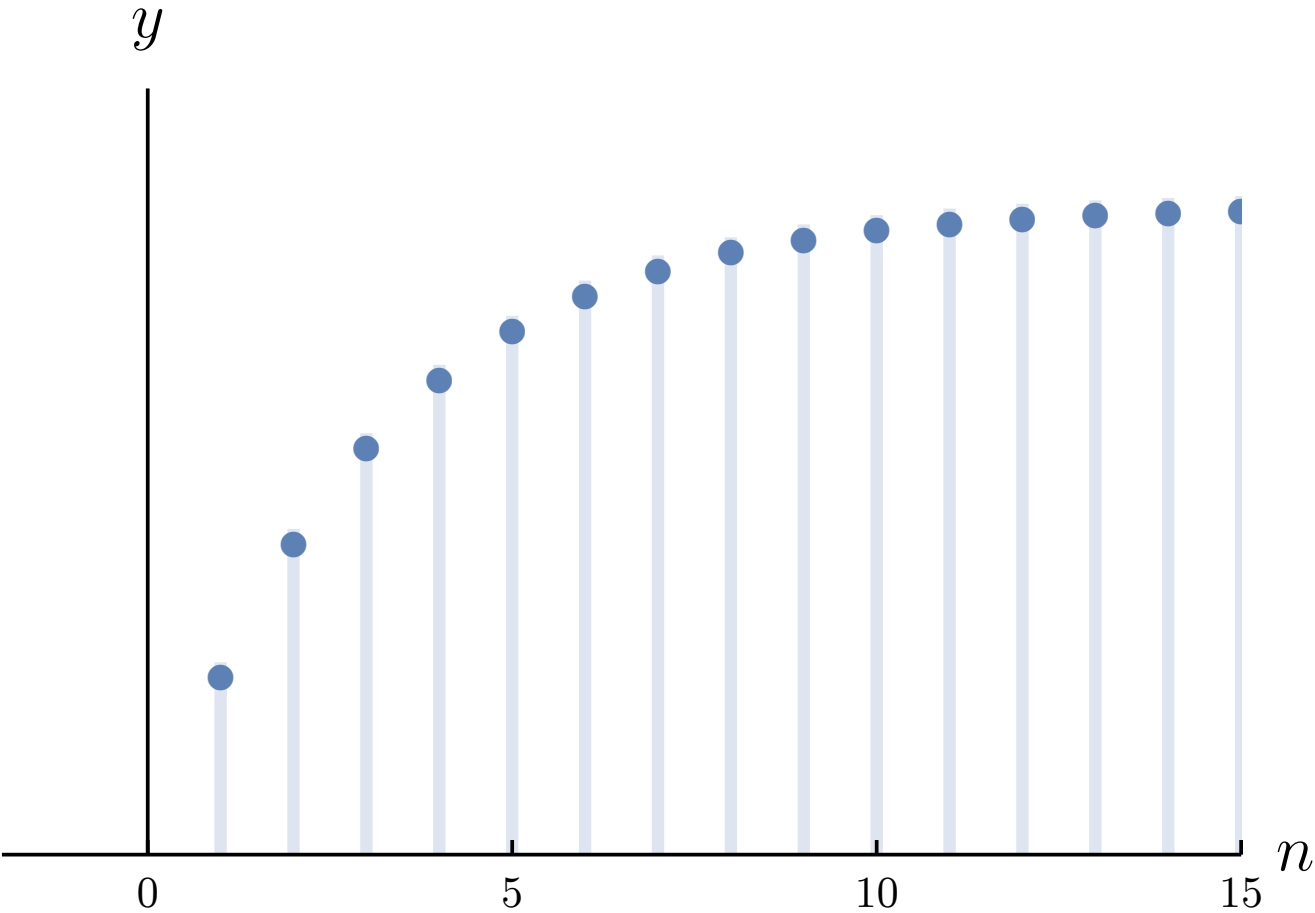
If

$$a_{n+1} < a_n$$

for all n then (a_n) is a **decreasing** sequence.

A sequence is **monotone** if it is either increasing or decreasing.

Increasing:



Definition #6 (Bounded Above and Below)

If $M \in \mathbf{R}$ satisfies

$$a_n \leq M$$

for all n then (a_n) is **bounded above** by M .

If m satisfies

$$a_n \geq m$$

for all n then (a_n) is **bounded below** by m .

We have

$$\frac{1}{n} < 5$$

and

$$\frac{(-1)^n}{n} > -2$$

for all n .

Definition #7 (Bounded)

If

$$m \leq a_n \leq M$$

for all $n \in \mathbf{N}$ then (a_n) is a **bounded** sequence.

Example #4

Determine whether the sequence is increasing, decreasing, or not monotone:

$$a_n = \frac{1 - n}{2 + n}$$

Here

$$a_n = \frac{1-n}{2+n} \Rightarrow a_{n+1} = \frac{1-(n+1)}{2+(n+1)} = \frac{-n}{3+n}$$

Now

$$\begin{aligned} a_n &> a_{n+1} \\ \frac{1-n}{2+n} &> \frac{-n}{3+n}, \end{aligned}$$

and with $n+3, n+2 > 0$

$$\begin{aligned} (3+n)(1-n) &> -n(2+n) \\ 3-2n-n^2 &> -2n-n^2 \\ 3 &> 0. \end{aligned}$$

Since $3 > 0$, it follows that $a_n > a_{n+1}$. This shows (a_n) is decreasing.

Example #5

Determine whether the sequence is increasing, decreasing, or not monotone:

$$b_n = 3 - 2ne^{-n}$$

Is the sequence bounded?

Let

$$f(x) = 3 - 2xe^{-x} \text{ then } f'(x) = (2x - 2)e^{-x}$$

Since $f'(x) > 0$ for $x > 1$ we know f and b_n are increasing.

As $n \rightarrow \infty$

$$\frac{n}{e^n} \rightarrow 0$$

by L'Hospital and so $b_n \rightarrow 3$ and is bounded below by b_1 , and above by 3.

Theorem #4 (Monotone Convergence Theorem)
If (a_n) is monotone and bounded then (a_n) converges.

This result is mainly used for technical arguments in later sections.

Intuitively, it makes sense. We are rising/falling to a limiting value.

